

Solutions to Linear Algebra Problems

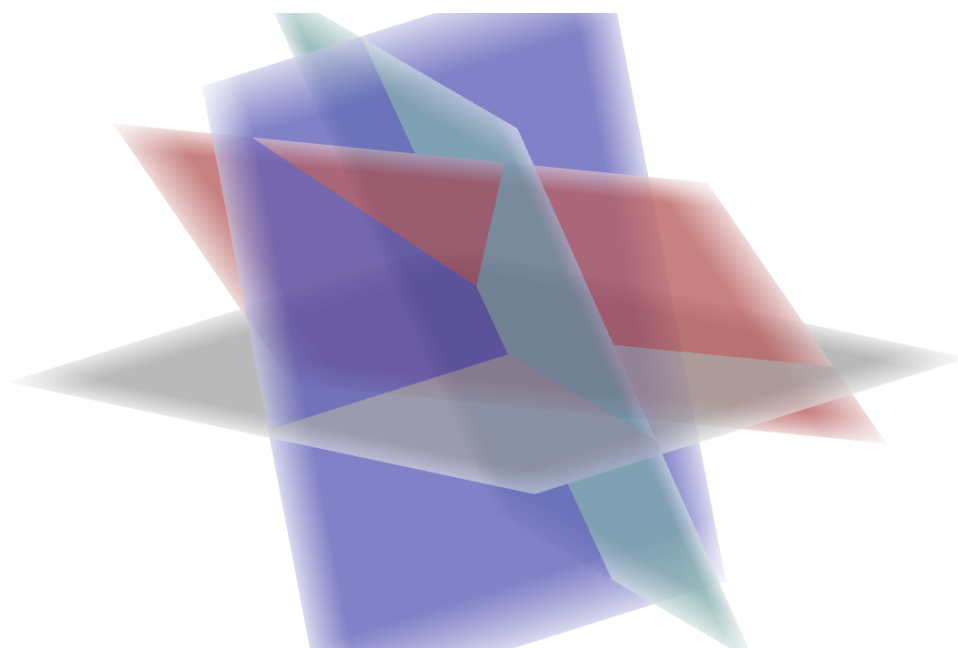
Minh Le

May 23, 2026

Abstract

This note collects my solutions to selected problems in Linear Algebra. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Linear Algebra Done Right 4e, Sheldon Axler



Problem 1.

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Source: *Linear Algebra Done Right* - Ex.12 p.25

Solution:

Let V_1, V_2 be two subspaces of V .

($\Rightarrow$): Suppose that $V_1 \cup V_2$ is a subspace of V . Suppose for contradiction that one of the subspaces V_1 or V_2 is not contained in the other. We can choose u, w such that $u \in V_1, u \notin V_2$ and $w \in V_2, w \notin V_1$.

Since $V_1 \cup V_2$ is a subspace, we have $u + w \in V_1 \cup V_2$. This means $u + w$ must be in V_1 or V_2 .

If $u + w \in V_1$, since V_1 is a subspace and $-u \in V_1$, then $w = (u + w) - u \in V_1$, which is a contradiction. Similarly, if $u + w \in V_2$, since V_2 is a subspace and $-w \in V_2$, then $u = (u + w) - w \in V_2$, which is also a contradiction.

Therefore, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

($\Leftarrow$): If $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$, then $V_1 \cup V_2 = V_2$ or $V_1 \cup V_2 = V_1$, which is a subspace.

Problem 2.

Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that the list $v_1, \dots, v_m$ is linearly independent if and only if the list $w_1, \dots, w_m$ is linearly independent.

Source: *Linear Algebra Done Right* - Ex.14 p.38

Solution:

($\Rightarrow$): Suppose the list $v_1, \dots, v_m$ is linearly independent. Consider a combination of $w_1, \dots, w_m$ such that:

$$\begin{aligned} c_1 w_1 + c_2 w_2 + \dots + c_m w_m &= 0 \\ \Rightarrow c_1 v_1 + c_2(v_1 + v_2) + \dots + c_m(v_1 + v_2 + \dots + v_m) &= 0 \\ \Rightarrow (c_1 + c_2 + \dots + c_m)v_1 + (c_2 + \dots + c_m)v_2 + \dots + c_m v_m &= 0 \end{aligned}$$

Since $v_1, \dots, v_m$ are linearly independent, it follows that $c_1 = c_2 = \dots = c_m = 0$. Therefore, the list $w_1, \dots, w_m$ are linearly independent.

($\Leftarrow$): Suppose $w_1, \dots, w_m$ are linearly independent. Consider a combination of $v_1, \dots, v_m$ such

that:

$$\begin{aligned}c_1v_1 + c_2v_2 + \cdots + c_mv_m &= 0 \\ \Rightarrow c_1w_1 + c_2(w_2 - w_1) + c_3(w_3 - w_2) + \cdots + c_m(w_m - w_{m-1}) &= 0 \\ \Rightarrow (c_1 - c_2)w_1 + (c_2 - c_3)w_2 + \cdots + (c_{m-1} - c_m)w_{m-1} + c_mw_m &= 0\end{aligned}$$

Since $w_1, \dots, w_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $v_1, \dots, v_m$ are linearly independent.

Problem 3.

Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Source: Linear Algebra Done Right - Ex.5 p.43

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of U and $w_1, w_2, \dots, w_n$ be a basis of W .

Since $V = U + W$, we have the list $\{u_1, u_2, \dots, u_m, w_1, w_2, \dots, w_n\}$ spans V .

Applying the procedure in Theorem 2.30 to reduce this list to a basis of V produces a basis consisting of the vectors $u_1, \dots, u_m$ and some of the w 's (none of the u 's get deleted in this procedure because $u_1, \dots, u_m$ is linearly independent).

Problem 4.

Suppose V is finite-dimensional and U is a subspace of V with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of V , each of dimension $n - 1$, whose intersection equals U .

Source: Linear Algebra Done Right - Ex.16 p.49

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of U . This list can be extended to a basis of V by adjoining $n - m$ additional vectors (by Theorem 2.32), say $w_1, w_2, \dots, w_{n-m}$. We denote this extended list as L .

Let V_i ($1 \leq i \leq n - m$) be the subspace spanned by L without including w_i . Clearly $\dim V_i = n - 1$ and $\bigcap_{i=1}^{n-m} V_i$ is also a subspace that contains U .

Consider $v \in \bigcap_{i=1}^{n-m} V_i$. We have:

$$\begin{aligned} a_1u_1 + a_2u_2 + \cdots + a_mu_m + b_2w_2 + b_3w_3 + \cdots \\ + b_{n-m}w_{n-m} = v \quad (\text{since } v \in V_1) \end{aligned}$$

$$\begin{aligned} c_1u_1 + c_2u_2 + \cdots + c_mu_m + d_1w_1 + d_3w_3 + d_4w_4 \\ + \cdots + d_{n-m}w_{n-m} = v \quad (\text{since } v \in V_2) \end{aligned}$$

By subtracting, we have:

$$\begin{aligned} (a_1 - c_1)u_1 + (a_2 - c_2)u_2 + \cdots + (a_m - c_m)u_m - d_1w_1 \\ + (b_2 - d_2)w_2 + b_3w_3 + \cdots + (b_{n-m} - d_{n-m})w_{n-m} = 0 \end{aligned}$$

Since L is a list of independent vectors and by repeating this process, this implies that all coefficients for w_i 's are 0 and v is uniquely represented in the basis of U .

We conclude that $\bigcap_{i=1}^{n-m} V_i = U$.

Problem 5.

Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

Source: *Linear Algebra Done Right* - Ex.13 p.58

Solution:

Let U be a subspace of V , let $S \in \mathcal{L}(U, W)$, and let $u_1, u_2, \dots, u_n$ be a basis of U . By Theorem 2.32, this list can be extended to a basis of V :

$$u_1, u_2, \dots, u_n, v_{n+1}, v_{n+2}, \dots, v_m.$$

For $1 \leq i \leq n$, let

$$w_i = Su_i.$$

Choose arbitrary vectors $w_{n+1}, w_{n+2}, \dots, w_m \in W$. By the linear map lemma, there exists a unique $T \in \mathcal{L}(V, W)$ such that

$$Tu_1 = w_1 = Su_1, \quad \dots, \quad Tu_n = w_n = Su_n,$$

and

$$Tv_{n+1} = w_{n+1}, \quad \dots, \quad Tv_m = w_m.$$

Now let $u \in U$. Since $u_1, \dots, u_n$ is a basis of U , there exist scalars $a_1, \dots, a_n$ such that

$$u = a_1u_1 + a_2u_2 + \dots + a_nu_n.$$

Thus

$$\begin{aligned} Tu &= T(a_1u_1 + a_2u_2 + \dots + a_nu_n) \\ &= a_1Tu_1 + a_2Tu_2 + \dots + a_nTu_n \\ &= a_1Su_1 + a_2Su_2 + \dots + a_nSu_n \\ &= S(a_1u_1 + a_2u_2 + \dots + a_nu_n) \\ &= Su. \end{aligned}$$

Therefore, $Tu = Su$ for all $u \in U$, so T is an extension of S to V .

Problem 6.

Suppose V and W are finite-dimensional and that U is a subspace of V . Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = U$ if and only if $\dim U \geq \dim V - \dim W$.

Source: Linear Algebra Done Right - Ex.18 p.67

Solution:

($\Rightarrow$) Suppose there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = U$. By the fundamental theorem of linear maps, we have:

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

$$\Rightarrow \dim U = \dim V - \dim \text{range } T \geq \dim V - \dim W$$

($\Leftarrow$) Suppose U is a subspace of V such that $\dim U \geq \dim V - \dim W$. Let $u_1, u_2, \dots, u_n$ be a basis of U . This list can be extended to $u_1, u_2, \dots, u_n, v_1, v_2, \dots, v_m$ to be a basis of $V \Rightarrow \dim V = n + m$. We have $\dim U \geq \dim V - \dim W \Rightarrow \dim W \geq \dim V - \dim U = m$. Therefore, we can let $w_1, w_2, \dots, w_m$ be an independent list of vectors in W . Define $T : V \rightarrow W$ as:

$$T(c_1u_1 + \dots + c_nu_n + d_1v_1 + \dots + d_mv_m) = d_1w_1 + d_2w_2 + \dots + d_mw_m$$

It is easy to verify that T is a linear map by checking $T(u + v) = T(u) + T(v)$ and $T(\lambda u) = \lambda T(u)$, $\forall u, v \in V$.

We have for all $u \in U$, $Tu = T(c_1u_1 + \dots + c_nu_n) = 0w_1 + \dots + 0w_m = 0 \Rightarrow u \in \text{null } T \Rightarrow U \subseteq \text{null } T$.

Now, let $u \in \text{null } T$. We have $Tu = 0 \Rightarrow d_1 = d_2 = \dots = d_m = 0$ since $w_1, w_2, \dots, w_m$ are

independent. Thus, u can be expressed as $c_1u_1 + c_2u_2 + \cdots + c_nu_n \Rightarrow u \in U \Rightarrow \text{null } T \subseteq U$.
Therefore, $U = \text{null } T$.

Problem 7.

Suppose there exists a linear map on V whose null space and range are both finite-dimensional. Prove that V is finite-dimensional.

Source: *Linear Algebra Done Right* - Ex.15 p.66

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of $\text{null } T$ and $v_1, v_2, \dots, v_n$ be a basis of $\text{range } T$. Let $w_1, w_2, \dots, w_n \in V$ such that $v_1 = Tw_1, v_2 = Tw_2, \dots, v_n = Tw_n$.

For any $z \in V$, we have:

$$\begin{aligned} Tz &= a_1Tw_1 + a_2Tw_2 + \cdots + a_nTw_n \\ &= T(a_1w_1 + a_2w_2 + \cdots + a_nw_n) \end{aligned}$$

Let $u = z - a_1w_1 - a_2w_2 - \cdots - a_nw_n \Rightarrow Tu = 0 \Rightarrow u \in \text{null } T$

$$\Rightarrow u = b_1u_1 + b_2u_2 + \cdots + b_mu_m$$

$$\Rightarrow z = a_1w_1 + a_2w_2 + \cdots + a_nw_n + b_1u_1 + b_2u_2 + \cdots + b_mu_m$$

This implies that the list $u_1, u_2, \dots, u_m, w_1, w_2, \dots, w_n$ spans V .

Therefore, V is finite dimensional.

Problem 8.

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row k , column k , equal 1 if $1 \leq k \leq \dim \text{range } T$.

Source: *Linear Algebra Done Right* - Ex.5 p.79

Solution:

By Thm 3.21: $\dim V = \dim \text{null } T + \dim \text{range } T$

Let $m = \dim \text{null } T = \dim V - \dim \text{range } T$

Let $u_1, u_2, \dots, u_m$ be a basis of $\text{null } T$. Since $\text{null } T$ is a subspace of V , this list can be extended to $v_1, v_2, \dots, v_n, u_1, \dots, u_m$ to be a basis of V . This implies $\dim V = m + n$ and $\dim \text{range } T = n$.

Consider $a_1Tv_1 + a_2Tv_2 + \cdots + a_nTv_n = 0$.

$$\Rightarrow T(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = 0$$

$$\Rightarrow a_1v_1 + \cdots + a_nv_n \in \text{null } T$$

$$\Rightarrow a_1v_1 + \cdots + a_nv_n = b_1u_1 + \cdots + b_mu_m$$

$$\Rightarrow a_1 = a_2 = \cdots = a_n = 0 \text{ since } v_1, v_2, \dots, v_n, u_1, \dots, u_m \text{ are independent}$$

This means $Tv_1, Tv_2, \dots, Tv_n$ are independent in W and this list can be extended to become a basis of W . With this construction of a basis of V and a basis of W , $\mathcal{M}(T)$ will have the desired entries.